Stage 2 Methodological Review: A/B/C/D Macro-Financial Forecasting

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Summary

The learning pipeline is sound, but the design over-promises on identification and power. B vs A can be judged on a 44-month historical test; C and D need a post-cutoff window, four control arms and live tracking well beyond the capstone before any claim is decisive.

The changes that matter most:

1. **Adopt a six-window split.** Burn-in Feb 2015–Jan 2019 (48 labels); development, already burned, Mar 2019–Dec 2022 (46 forecasts); primary historical test Jan 2023–Aug 2026 (44); full-span secondary Mar 2019–Aug 2026 (90); an LLM post-cutoff window; live forecasts from Nov 30, 2026 (Section 3).
2. **Treat pre-cutoff C and D results as non-evidence.** Memorization makes them unidentifiable. Primary C/D evidence is the post-cutoff window plus the prospective record (Sections 3 and 7).
3. **Add control arms.** A′ (mechanical regime labels), Cπ (permuted labels), C0 (no retrieval) and D0 (random graph) separate reasoning from flexibility, memory and retrieval volume. Add a trend control to B (Sections 5 and 6).
4. **Fix point-in-time details.** Train on labels through t−2, winsorize surprises at ±3, index releases by actual timestamp with rules for the 2025–26 shutdown gaps, and build returns from ex-date total-return series (Section 2).
5. **Pre-register one pooled test per contrast and a five-part decision rule.** At a realistic 1–2% R^2, the 44-month test has only 14–21% power, so report estimates with intervals first (Sections 4 and 9).
6. **Fix the portfolio rule now.** Run the benchmark's naive forecasts through the same optimizer, so portfolio differences come from forecasts alone (Section 8).

Keep 48 labels as the training floor, and keep standard RAG as C. Both are defensible with the changes above.

1. What is strong

The skeleton is sound: a common baseline, one downstream learner, identical dates, and real-time feature construction are exactly what an incremental-value study needs.

- **A nested ladder with one learner.** Every system feeds the same per-asset Ridge, so differences come from features, not model class. This is the right design for incremental-value claims.
- **Real-time feature construction.** Standardizing surprises on the prior 12 releases, training-fold-only scaling, expanding refits and inner expanding CV all avoid the most common look-ahead leaks.
- **A one-day execution lag.** A month-end evidence cutoff with execution at the next close removes same-close timing leaks (for example, VIX settles at 4:15 pm, after the 4:00 pm ETF close).
- **An intercept-only option in the Ridge grid.** It lets each model fall back to the historical mean, the benchmark that most predictors fail to beat out of sample (Welch & Goyal 2008; Campbell & Thompson 2008).
- **The right evidence hierarchy.** Forecast accuracy is primary and portfolio results secondary. Portfolio outcomes add many extra degrees of freedom (weights, costs, constraints).
- **Categorical agent outputs, not pseudo-probabilities.** Supported / Contradicted / Insufficient interacted with quantitative signals models state-dependent news effects, which the literature documents (Boyd, Hu & Jagannathan 2005; Campbell, Pflueger & Viceira 2020).
- **Holding C and D to one agent, prompt, model and schema.** That leaves retrieval as the intended treatment.
- **Honest contamination labels.** Calling 2019–2022 burned and later history pseudo-OOS is correct, and freezing plus hashing makes the claims auditable.
- **Forums used for hypotheses only.** That is the right epistemic role for practitioner communities.

2. What is weak, confounded or biased

The largest problems are not in the Ridge pipeline. They sit in data timing, feature scaling, agent contamination and the treatment definitions. Each row below has a concrete fix.

Issue	Why it matters	Fix
A's economic premise is thin	Release surprises are priced within minutes (Andersen et al. 2003). By the next month's first close the reaction is in the price, so A works only through slow drift or as a proxy for the macro state. Expect A's OOS R^2 near zero.	Acceptable for incremental tests, but report A vs the benchmark first and say so. Make growth and inflation block composites the primary A features (fewer, less noisy inputs).
12-release scaling is fragile	COVID-era releases yield standardized values in the tens, then inflate the trailing scale for a year. A few points then dominate Ridge coefficients in every later refit.	Winsorize standardized surprises at ±3 (frozen). Robust scale (median absolute deviation) and a 24-release window go to robustness.
Release disruptions inside the test window	October 2025 CPI was never published; October payrolls appeared with November on Dec 16, 2025 (Reuters). October PPI and retail sales slipped (Calculated Risk). January 2026 jobs and CPI moved to Feb 11 and Feb 13 (Chase).	Index each surprise by its actual release timestamp. Treat canceled releases as missing. Count the 12-window in releases, not months. Use first-print actuals only.
Label-completion lag	At month-end t, the label for decision t−1 ends at the first close of t+1, so it is still unknown. Using it leaks a month of returns. The same leak can hide inside inner CV folds.	Train on labels through t−2. Purge one month between each inner training fold and its validation month.

Issue	Why it matters	Fix
Distribution timing at the execution boundary	IEF and LQD go ex-dividend on the first business day of the month, which is the execution day (iShares schedule). A misdated LQD distribution (about 0.4% of price) moves a month's income between labels, and December 2025 had two (Dividend Vision).	Build returns from ex-date total-return indices. Unit-test every month against Bloomberg total return.
Contaminated agent training data	Before the model's training cutoff, agent labels can encode memorized outcomes. Instructions to respect dates and masking both fail to stop recall (Lopez-Lira, Tang & Zhu 2025). Ridge learns those weights and carries them into post-cutoff forecasts.	Treat pre-cutoff C/D results as non-evidence. Run a pre/post-cutoff break test on agent-feature informativeness (Sections 3 and 7).
Causal graph built with hindsight	A graph drafted in 2025–26 can encode lessons from 2021–23 (supply shocks, rapid hikes, the stock–bond correlation flip). D then carries hindsight that C lacks.	Freeze, date and hash the graph. Derive edges from pre-2015 literature or theory, and log every edit.
Interaction coding	Three one-hot interactions sum to the raw signal, so they overlap A's main effect and the Ridge penalty depends on the coding.	Keep A's main effect and add only Supported and Contradicted interactions, with Insufficient as the reference. Shrinkage then pulls C and D back toward A.

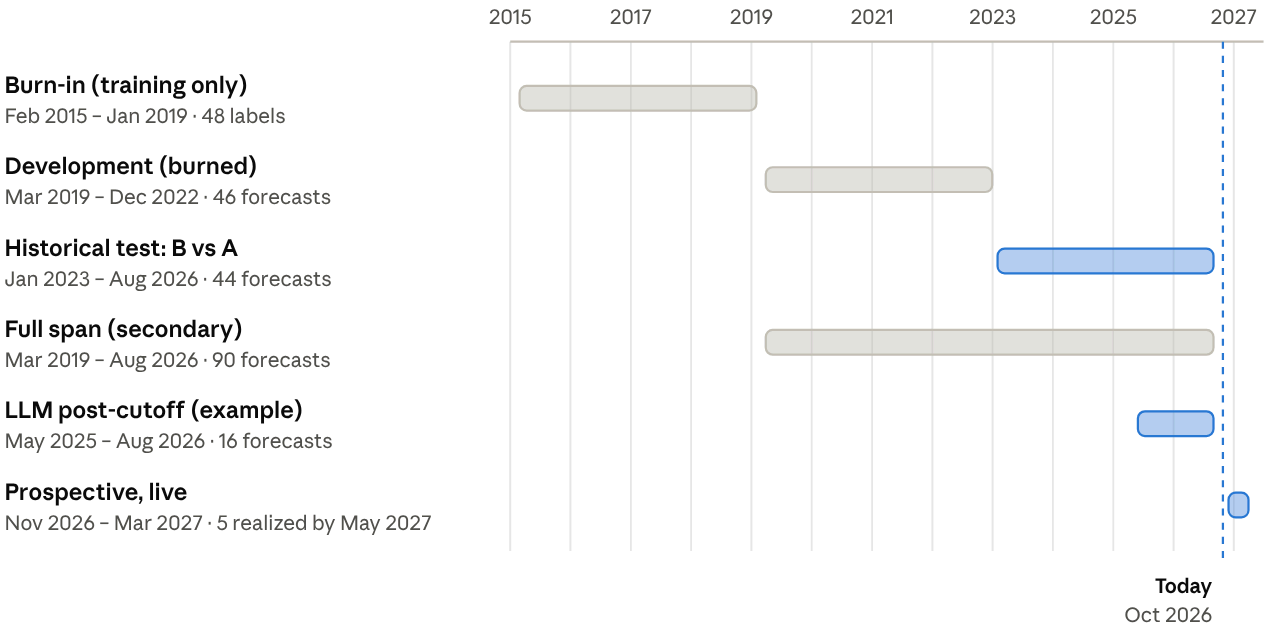
Issue	Why it matters	Fix
No forecast-to-weight rule	"Long-only, fully invested" leaves the most consequential portfolio choice open, which invites post-hoc selection.	Pre-register one rule and a static benchmark portfolio (Section 8).
Benchmark window	If the historical mean uses a different window from the Ridge intercept, A vs benchmark mixes two effects.	Compute the benchmark on the same expanding window, so intercept-only Ridge equals it exactly.

3. Recommended sample structure

Use six windows, each tied to the one claim it can support: a 48-label burn-in, the burned 2019–2022 period as development, January 2023 to August 2026 as the primary historical test, a full-span secondary, a post-cutoff window for C and D, and live forecasts from November 30, 2026. All dates below come from the NYSE calendar under your execution convention.

B is judged on 44 post-2022 months; C and D on short post-cutoff and live windows

Decision month-ends, drawn to scale. Blue marks the primary evidence windows.



Sample windows · NYSE calendar via exchange_calendars · post-cutoff window shown for an example model cutoff of January 2025

The primary test begins where the burned development returns end, on February 1, 2023. The LLM windows are short, so C and D rest on the least data.

Window	Decision month-ends	Holding periods	Size	Use	Claim it supports
Burn-in	Feb 2015 – Jan 2019	Mar 2, 2015 – Mar 1, 2019	48 labels	Initial estimation only	None
Development (burned)	Mar 2019 – Dec 2022	Apr 1, 2019 – Feb 1, 2023	46 forecasts	Remaining tuning, prompt and graph development, debugging, COVID and 2022 diagnostics	Development results
Historical test (primary for B vs A)	Jan 2023 – Aug 2026	Feb 1, 2023 – Oct 1, 2026	44 forecasts	One run after the freeze	Historical pseudo-OOS

Window	Decision month-ends	Holding periods	Size	Use	Claim it supports
Full span (pre-registered secondary)	Mar 2019 – Aug 2026	Apr 1, 2019 – Oct 1, 2026	90 forecasts	Power and regime coverage	Pseudo-OOS, partly selection-contaminated
LLM post-cutoff (primary historical evidence for C and D)	First month-end at least 3 months past the model's cutoff – Aug 2026	From the month after	16 if the cutoff is Jan 2025	C vs A and D vs C	Post-cutoff historical
Prospective	Nov 30, 2026 onward	From Dec 1, 2026	4–5 realized by May 2027	All systems, live	Cleanest evidence, underpowered for years

How the boundaries are derived

1. **The first forecast is March 29, 2019, not February.** Decision m's label runs from the first close of m+1 to the first close of m+2. At month-end t only labels through t−2 are complete. The 48th label (January 2019 decision) completes on March 1, 2019; the February 2019 decision would have only 47.
2. **The last realized decision is August 31, 2026.** Its holding period runs September 1 to October 1, 2026. The September 30 decision does not close until November 2, 2026. Confirm this in the frozen data pull, not the calendar: exit-day prices, the October 1 IEF and LQD distributions and the T-bill accrual must all be present and final.
3. **The dev/test boundary is the January 2023 decision.** The December 2022 return ends February 1, 2023, exactly where the January 2023 return starts, so no return overlaps. If your 2019–2022 comparison was indexed by return month, the December 2022 decision was never used; keep the conservative boundary anyway. The first test forecast trains on labels through November 2022, so COVID and the 2022 shock are in its training set.
4. **2022 stays out of the test.** It was used for selection, and choosing a split after looking is the split-point mining that produces large size distortions, worst with short evaluation samples (Hansen & Timmermann 2012). The rate shock is still evaluated, in development diagnostics and in the full-span secondary.
5. **Starting the test later would not make it cleaner.** Later data were already inspected, so a later start purifies nothing and cuts power, which is strongest with long evaluation periods (same source).

6. **The full span earns a pre-registered secondary slot.** The 2019–2022 selection was one binary choice about A's architecture, so its selection bias is small, while 90 forecasts roughly double the sample.
7. **Know what the test lacks.** It covers the March 2023 bank stress, the 2023 long-end selloff, the 2024–25 easing, the April 2025 tariff shock and the 2025–26 data disruptions. It has no recession and only the last four hikes of the 2022–23 cycle, so results say little about those regimes.

The LLM post-cutoff window

- **Set the boundary empirically.** Use the later of the documented training-data cutoff plus 3 months and the last month the model recalls in a no-retrieval memorization probe plus 3 months. Probe with exact values of CPI, payrolls and month-end ETF prices around the cutoff, the test design in Lopez-Lira, Tang & Zhu (2025).
- **Choose the model with the window in mind.** A model whose knowledge runs into mid-2026 leaves no post-cutoff history at all. An older pinned snapshot trades capability for a longer clean window. Pin the dated snapshot and keep it through the prospective period.
- **Add a vintage-consistent robustness arm.** Chronologically consistent models are trained only on text available before a fixed date: ChronoGPT and its instruction-tuned version (He et al. 2025a; 2025b) or DatedGPT with annual cutoffs from 2013 to 2024 (Yan et al. 2026). Matching each test year to an earlier vintage gives a leakage-free C vs D over the whole 2023–26 test, with a weaker model.
- **State the remaining limit.** Post-cutoff C and D forecasts still use Ridge weights learned on pre-cutoff agent labels. Section 7 gives the break test that measures this.

The prospective window

- **Freeze before November 30, 2026.** Hash and timestamp code, configuration, prompts, the causal graph, the model snapshot ID and the retrieval index, and register them on OSF or in a public timestamped commit. If you can freeze by October 30, start a month earlier.
- **Expect 4–5 labels by spring.** Decisions on November 30, December 31, January 29, February 26 and March 31 exit on January 4, February 1, March 1, April 1 and May 3, 2027.
- **Treat it as an integrity check, not a significance test.** Its job is to show the pipeline ran in real time with outputs logged before outcomes. Pre-register a continuation with BNY of at least 36 months with a fixed analysis date.

- **Get more evidence per month.** Score each mechanism label against the next release of its own indicators, for example whether Inflation: Supported precedes positive CPI surprises. That yields up to eight release-level outcomes a month and tests the agent's reasoning directly.

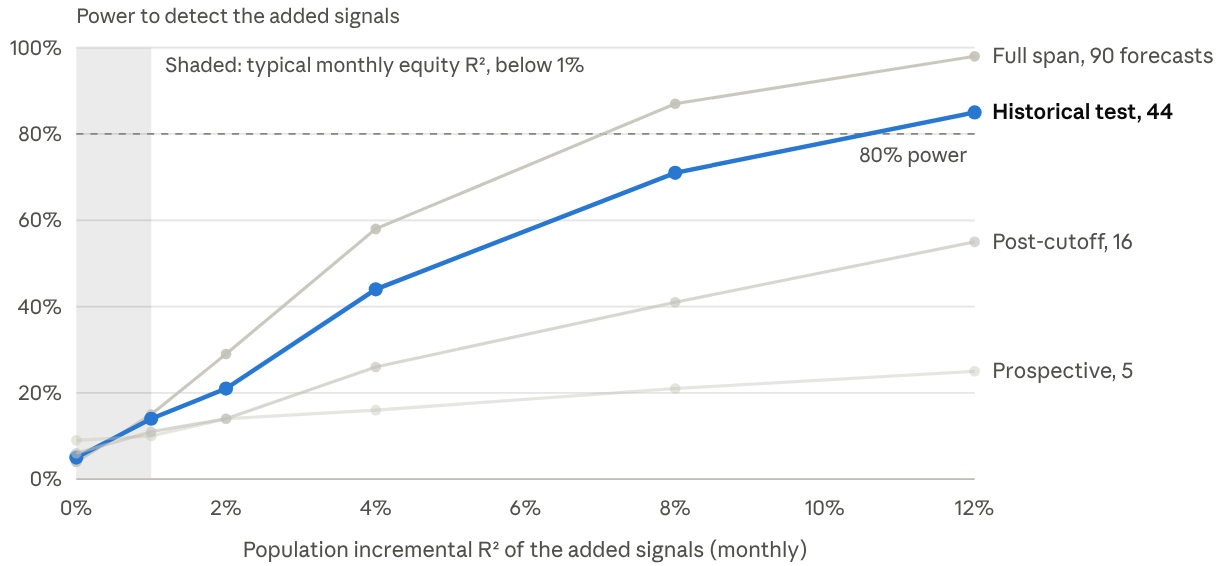
4. Is 48 months enough training?

Yes, as a floor, provided you cut the feature count. The binding constraint is evaluation power, not training length.

- **Parameter count.** A has 8 features, B up to 13 with the Section 5 signals, and C or D up to 14 with the coding in Section 2. At 48 labels that is up to 0.29 parameters per observation, workable only with heavy shrinkage.
- **Inner CV is the weak link.** With a 36-label inner minimum and a one-month purge, the first refits pick alpha from about 11 validation errors. A broad grid on so few points is close to random, so expect intercept-only to win early. That is honest, not a bug.
- **A longer burn-in costs test months.** Each extra training month removes a forecast, and power rises with evaluation length (Hansen & Timmermann 2012). Out-of-sample tests already have less power than in-sample ones (Inoue & Kilian 2005).
- **Recommendation.** Freeze 48 as the minimum. Build the C and D interactions on two block composites (growth: payrolls, retail sales, IP, ISM; inflation: CPI, core CPI, PPI, AHE) rather than eight indicators. Use about 9 log-spaced alphas plus intercept-only, and run 36 and 60 as robustness.

Even pooled across assets, 44 test months reach 80% power only near a 10% R²

Simulated power of a one-sided 5% Clark–West test, by evaluation window



Simulation run for this review · 800 paths per point · t(5) errors, AR(1) predictor, 3 signals with 1 relevant, expanding OLS, pooled over SPY/IEF/LQD with error correlations from −0.1 to 0.7

Monthly equity-premium predictors rarely exceed about 1% out-of-sample R² (Campbell & Thompson 2008). At 1–2%, the 44-month test detects an added signal 14–21% of the time even pooled across assets. The 16- and 5-month windows also over-reject when nothing works (6–9% against a nominal 5%). Report estimates with intervals and pre-registered decision rules (Section 9), not pass/fail significance.

5. Conventional signals for System B

All three are defensible because each has pre-2015 academic support, so 2015–2026 is genuinely outside their publication samples. Two need redefinition, and B needs a trend control. Expect smaller effects than published, since predictors decay after publication (McLean & Pontiff 2016).

Signal	Verdict	Evidence and problem	Change
Credit: 3-month change in BAA–10Y	Keep, add the level	Spread levels track expected returns on stocks and bonds (Fama & French 1989), and credit-spread information predicts activity (Gilchrist & Zakrajšek 2012). A 3-month change is a deterioration signal with an ambiguous sign; for LQD the level is the carry. Moody's Baa index targets maturities near 30 years, so BAA–10Y also carries the 10s–30s curve.	Use the standardized level and the 3-month change, with signs pre-registered. Robustness: Bloomberg IG index OAS, closer to LQD's universe. Avoid FRED's ICE BofA OAS, which keeps only 3 years of history from April 2026 (FRED). Avoid the excess bond premium as a real-time input: it is a fitted regression residual whose history can change with each update.
Treasury curve: 10Y–3M	Keep	It is the classic recession predictor (Estrella & Mishkin 1998), and the term spread predicts bond excess returns (Campbell & Shiller 1991). It is highly persistent, and it stayed inverted from late 2022 to late 2024 without a recession, inside your test window. Its 3M leg nearly equals the risk-free rate in the target.	Keep as primary. Robustness: 10Y–2Y. Pre-register the expected sign for each asset.

Signal	Verdict	Evidence and problem	Change
Variance risk premium: VIX^2 minus trailing 22-day RV	Keep, fix units and timing	The premium predicts equity returns, most strongly at quarterly horizons, and the result depends on model-free implied variance and high-frequency realized variance (Bollerslev, Tauchen & Zhou 2009). Daily close-to-close RV is noisier, and VIX settles 15 minutes after the ETF close.	Put both legs in monthly percent^2: $VIX^2/12$ minus the sum of 22 squared daily S&P 500 log returns ($\times 10^4$). Use same-time closes. Robustness: the expected-variance version of Bekaert & Hoerova (2014).
Add: own-asset 12-month excess return (time-series momentum)	Add as a control	Trend is the most robust conventional signal across equity indices and bond futures (Moskowitz, Ooi & Pedersen 2012). Retrieved news is full of recent price action, so an agent can look skilful by restating the trend.	Add it to B, and test C and D against B as well as A (Section 6).

Two cross-cutting rules apply. Pre-register each signal's expected sign per asset, and run a sign-restricted Ridge as robustness; sign restrictions improve out-of-sample equity-premium forecasts (Campbell & Thompson 2008). Keep the forum-sourced implementation choices, such as the 3-month window, on a written list, because they are where snooping enters.

6. Do A/B/C/D isolate the claimed effects?

B vs A does; C vs A and D vs C do not, because each bundles several treatments. Four cheap control arms and one encompassing test separate them. Three of the four need no extra LLM calls.

Contrast	What actually changes	Confounds	What isolates the claim
B vs A	Five market-signal features (Section 5)	None material; re-tuned alpha is part of the treatment	Nothing extra
C vs A	Retrieved text, the LLM's parametric knowledge (including memorized outcomes), extra interaction parameters, and whatever signal the policy mechanism uses	Added flexibility alone can help or hurt; memory can masquerade as reasoning; a market-based policy signal smuggles B into C	**A′**: the same one-hot interactions with mechanical labels from A's own data (for example, the sign of the trailing three-release composite). **Cπ**: C's labels permuted across months, preserving their frequencies. **C0**: a no-retrieval agent, used as a contamination diagnostic.
D vs C	The causal graph, but also which documents arrive, how many, how long, how recent, and how the query is phrased	Any retrieval difference, plus hindsight in the graph	**D0**: the same graph-guided retrieval over a randomly rewired graph of equal size and degree distribution. Hold corpus, index, embedding model, date filter, document count and token budget fixed across C, D and D0.
C or D vs B (not claimed, but BNY will ask)	Agent signals vs cheap market signals	Retrieved text restates spreads, curve and trend	A forecast-encompassing test of C on B (Harvey, Leybourne & Newbold 1998), or a B+C arm

The claims then read cleanly. C beats A′ means the agent adds judgment beyond a mechanical regime rule. C beats Cπ means its labels carry information beyond added parameters. D beats D0 means causal structure, not graph-shaped retrieval, does the work.

Three further rules matter:

- **Keep the policy mechanism on A-only data.** Interact Policy transmission with the inflation composite, not the curve or a rate change. Otherwise C vs A partly measures B.
- **Measure label disagreement before looking at returns.** D and C can differ only in months where their labels differ. At 85% agreement, a 44-month test compares about 7 months. Report agreement and Cohen's kappa per mechanism, and pre-register a disagreement floor below which D vs C is declared unidentified.
- **Control sampling noise.** Run each prompt five times with fixed seeds, use the modal label, report run-to-run agreement, and use identical seeds for C, D and D0.

7. Is standard RAG vs causal RAG the cleanest C-vs-D design?

Yes, timestamped RAG is the right C. A no-retrieval C would change two things at once: whether the agent sees evidence, and how that evidence is organized. It is clean only with matched retrieval, a random-graph placebo, a hindsight-free graph and contamination diagnostics.

- **It matches how causal RAG is evaluated.** CausalRAG was benchmarked against regular RAG and graph-based RAG (Wang et al. 2025). Your D0 random-graph arm plays the graph-RAG role: same machinery, no causal content.
- **Keep the no-retrieval agent, as a diagnostic.** If C0 matches C before the cutoff but collapses after it, the pre-cutoff skill was memory, not evidence.
- **Contamination may not be symmetric.** Graph prompts that ask what happened downstream of an event may invite more recall than plain retrieval. The bias can also run the other way: in-sample, general knowledge hurt GPT sentiment signals more than look-ahead helped (Glasserman & Lin 2023). Pre-cutoff D vs C is therefore not just inflated; its sign is unreliable.
- **Run a cutoff break test.** For C and D separately, estimate the agent features' contribution with a post-cutoff interaction. A significant drop at the cutoff is the signature of memorization.

- **Use anonymization as robustness, not as the cure.** Masking entities and dates alone fails, because models reconstruct them from context, while rewriting text into anonymized economic logic reduces recall (Lopez-Lira, Tang & Zhu 2025). Run an anonymized-corpus variant of C and D.
- **Protect the corpus timestamps.** Use first-publication time, drop documents edited after publication, and use one embedding model for C, D and D0. Memorization extends to embeddings (same source), so retrieval ranking itself can leak.
- **Build the graph without hindsight.** Take nodes and edges from pre-2015 macro-finance literature, cite each edge, freeze and hash it before any C or D run, and make D0 by degree-preserving rewiring of that graph.

8. Frozen parameters vs robustness checks

Freeze everything that defines the estimand, the information set and the decision rule; vary only implementation details, and never let a robustness result replace the primary one. Robustness results are all reported, none selected.

Item	Frozen primary	Robustness only
Data	Sources, first-print actuals, Bloomberg consensus median at release, pull date and file hashes	None
Windows	The six windows in Section 3 and the order they are opened	Excluding March–June 2020 from the full span; excluding October 2025–February 2026 release months
Target	Excess total return from ex-date-adjusted total-return indices, minus the 1-month T-bill accrued over exact holding days	3-month bill or SOFR as the risk-free rate
Timing	Month-end evidence cutoff; execute at the first trading day's close; hold to the next execution	Execute at the month-end close or the second trading day's close

Item	Frozen primary	Robustness only
Macro features	8 surprises scaled on the prior 12 releases, winsorized at ±3; growth and inflation composites for interactions; missing set to 0 after training scaling	24-release window, median-absolute-deviation scale, winsorizing at ±2 or ±4, eight indicator-level interactions
B signals	Credit level and 3-month change, 10Y–3M, VRP in monthly percent², 12-month trend	IG index OAS, 10Y–2Y, expected-variance VRP
Model	One Ridge per asset, expanding window, 48-label minimum, monthly refit, labels through t−2	36 or 60 labels minimum; 60-month rolling window; sign-restricted Ridge
Tuning	Inner expanding CV, 36-label inner minimum, one-month purge, MSE loss, 9 log-spaced alphas plus intercept-only, ties to the larger alpha	One fixed alpha chosen on the development window; finer grid
Agent	Pinned model snapshot, prompt text, schema, seeds, five runs with the modal label, corpus, index, embedding model, document count, token budget, hashed graph	Anonymized corpus; chronologically consistent model; document count ±50%
Interaction coding	A main effect plus Supported and Contradicted interactions, Insufficient as reference	Full one-hot coding
Portfolio	Long-only, fully invested mean-variance with risk aversion 5 and a 36-month Ledoit–Wolf shrunk covariance. The benchmark portfolio is the same optimizer fed historical-mean forecasts.	Risk aversion 2 and 10; a rank-based tilt around equal weights
Costs	5 bps one-way on turnover	0, 2, 10 and 25 bps
Inference	The tests, bootstrap, significance level, multiplicity rule and decision rules in Section 9	Alternative block lengths

The portfolio rule matters most. Feeding the benchmark's naive forecasts through the same optimizer means any portfolio difference comes from the forecasts alone.

9. Statistical evaluation framework

Decide each contrast with one pre-registered test pooled across assets, Holm-adjusted across the three primary contrasts, and claim value only if economic and per-asset checks agree. Given the power in Section 4, report every result as an estimate with a 90% interval first and a test second.

Forecast accuracy (primary)

The headline metric is out-of-sample R^2 of the larger model against the smaller one, per asset (Campbell & Thompson 2008). For nested pairs, the test uses the Clark–West adjusted loss differential, which corrects the noise a larger nested model adds (Clark & West 2007).

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R^2_{OOS} = 1 - \frac{\sum_t \left(r_{t+1} - \hat r^{U}_{t+1}\right)^2}
{\sum_t \left(r_{t+1} - \hat r^{R}_{t+1}\right)^2}, \qquad f_{t+1} =
\left(r_{t+1} - \hat r^{R}_{t+1}\right)^2 - \left[\left(r_{t+1} - \hat
r^{U}_{t+1}\right)^2 - \left(\hat r^{R}_{t+1} - \hat
r^{U}_{t+1}\right)^2\right]
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Comparison	Type	Test
A vs historical mean; B vs A; C vs A; D vs A	Nested	One-sided Clark–West
D vs C; C vs A′; D vs D0	Non-nested	One-sided Diebold–Mariano with the small-sample correction (Diebold & Mariano 1995; Harvey, Leybourne & Newbold 1997)
C vs Cπ	Randomization	Share of 200 label permutations whose MSPE gain is at least C's; Ridge reruns only, no LLM calls
C or D vs B	Encompassing	Forecast-encompassing regression of B's error on the C − B forecast difference

- **Pool across assets.** Scale each asset's errors by its expanding-window return standard deviation, so SPY does not dominate, then test the cross-asset average differential. This is the primary statistic for each contrast; per-asset tests are secondary.
- **Bootstrap jointly.** Use the stationary bootstrap (Politis & Romano 1994) with automatic block length (Politis & White 2004), resampling the three assets by month together to keep their correlation.
- **Control multiplicity.** Apply Holm at 5% across the three primary pooled contrasts (B vs A, C vs A, D vs C). Label everything else exploratory, and summarize all systems with a model confidence set (Hansen, Lunde & Nason 2011).
- **Show when gains occur.** Plot cumulative squared-error differences (Welch & Goyal 2008) and run the fluctuation test (Giacomini & Rossi 2010), so a single episode cannot pass as persistent skill.
- **Add a harder bond benchmark.** For IEF and LQD, also report results against a carry forecast (current yield minus the T-bill rate). The historical mean is an easy bar for bonds.

Portfolio performance (secondary)

- **Certainty-equivalent gain** over the benchmark portfolio, net of 5 bps, at the optimizer's risk aversion (Campbell & Thompson 2008).
- **Sharpe ratio difference** with the robust bootstrap test of Ledoit & Wolf (2008), plus turnover and the break-even cost.
- **Deflated Sharpe ratio** (Bailey & López de Prado 2014), using a log of every configuration the team has run. Start that log now; it also feeds the probability of backtest overfitting (Bailey et al. 2017).
- **Directional accuracy** with the Pesaran–Timmermann (1992) test, as description only.

Decision rule (pre-register verbatim)

A system adds incremental value only if all of these hold in its designated window (the historical test for B; the post-cutoff window for C and D):

1. The pooled one-sided test rejects at 5% after Holm.
2. Out-of-sample R^2 is positive for at least two of the three assets.
3. The certainty-equivalent gain is positive net of costs.
4. For C, it also beats A′ and Cπ; for D, it also beats D0.
5. For C and D, the prospective point estimate has the same sign.

Otherwise the verdict is "inconclusive", reported with the point estimate, its 90% interval and the minimum detectable effect from Section 4.

10. Five biggest validity risks

Ranked by how likely each is to overturn a conclusion. The first two decide what can be claimed at all; the other three decide whether a claimed effect is real.

Rank	Risk	How it breaks the conclusion	Guard
1	LLM memorization of outcomes	Pre-cutoff C and D skill is not identified: memory and reasoning give the same outputs (Lopez-Lira, Tang & Zhu 2025). Contaminated Ridge weights carry into post-cutoff forecasts, and the bias can differ between C and D.	Post-cutoff window as primary C/D evidence; memorization probe; cutoff break test; C0 diagnostic; a chronologically consistent model arm
2	Too little data for the question	At a realistic 1–2% R^2, the 44-month test detects an added signal 14–21% of the time, and 5 prospective months cannot test anything. Both "no value" and a lucky "value" are likely errors.	Estimation framing with intervals; pooled tests; minimum detectable effect stated up front; 36+ months of continued live tracking
3	Researcher degrees of freedom	2019–2022 is burned, later results were inspected, and prompts, graph and signal tweaks can all be iterated until something works. Choosing among many paths inflates every test (Harvey, Liu & Zhu 2016).	Freeze before opening the test; a log of every configuration run; deflated Sharpe ratio; Holm across primary contrasts; no change after unblinding

Rank	Risk	How it breaks the conclusion	Guard
4	Confounded treatments	C vs A can reflect extra parameters or memorized facts rather than reasoning; D vs C can reflect retrieval volume, recency or graph hindsight rather than causal structure; high label agreement leaves few informative months.	A′, Cπ and D0 arms; matched retrieval budgets; pre-2015 graph; a pre-registered disagreement floor
5	Point-in-time data errors	Revised actuals, misaligned release dates, the 2025–26 cancellations, consensus timing, misdated distributions on execution days, edited documents and COVID outliers each leak information or distort labels.	First-print data; release-timestamp indexing; an audit table for every October 2025–February 2026 release; total-return reconciliation; ±3 winsorizing; first-publication document timestamps

References

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Data and release facts

- Reuters via Yahoo News: October 2025 CPI canceled; October payrolls with November on Dec 16
- Calculated Risk: October 2025 retail sales and PPI not released on Nov 14
- Chase: January 2026 partial shutdown and rescheduled jobs and CPI dates
- iShares 2024–2025 distribution schedule
- Dividend Vision: LQD distribution history
- FRED: ICE BofA BBB OAS series note on 3-year history
- Split dates: NYSE calendar via the `exchange_calendars` Python package. Power figures: simulation run for this review.